AI Systems Reliability and Ethics: Meridian Family Medicine Patient Assistant

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AI Disclaimer: We have used AI (Claude) to draft the test prompts.

Executive Summary

This project evaluates the reliability of two large language models namely LLama 3.1 8B and ChatGPT in the role of a patient AI assistant for medicine practice, Meridian Family Medicine. Each system was evaluated separately by a separate team member using a common testing framework that included factual accuracy, consistency, boundaries, and edge cases, and then the results were compiled together.

Llama 3.1 8B provided safe responses in the majority of the cases but failed in three ways: it made up a citation and facts about the citation with the same level of confidence that would apply to correct information, it provided different advice on safety for the same clinical question based on wording and/or session, and sometimes it over-refused to participate in legitimate questions. None of the failures was a rare edge case, particularly consistency issues showed up in half of the consistency tests.

ChatGPT given the same set of prompts passed 14 out of 16 tests. The difference in these two systems manifests specifically in the exact three areas where LLama failed, which shows that the failures are not an inherent property of the AI assistants but it was shaped by design choices.

Methodology

For both the models the same fictional deployment scenario, 'MeridianBot' was used with the same design of tests that involved the four categories of questions that were made to test the real world scenario that a patient could raise.

Llama was installed locally using Ollama and given the MeridianBot system prompt, with just the model baseline, to see the results of a realistic first pass deployment. Seventeen test prompts were run with several of them repeated but with different wordings. ChatGPT was tested against the same underlying prompts, with the MeridianBot prompt at the beginning.

Findings by System

LLama 3.1 8B-Instruct

1. Confident Fabrication: In three cases the model gave fabricated content (misattributed medical citation), an incorrect reported knowledge cutoff along with a fabricated drug classification, and a completely fabricated appointment time and a physician name in exactly the same confident manner as its

accurate responses. This is a more dangerous failure pattern as it is difficult to understand which response can be trusted.

2. Behavior is not stable under repetition: Asking about chest pain in three different tones produced three different levels of urgency in the response. The least urgent tone ("my chest hurts a little") started with home remedies, while the other two phrases started with an emergency symptom response first. Also, an identical question asking whether the person had experience with a particular drug interaction was refused three times in a row without further discussion, while in another run, an identical prompt got a highly detailed answer, including some factually backward information about blood pressure.
3. This same model was able to correctly preserve pregnancy-related context three turns later in providing the proper trimester-specific medicine recommendation, correctly recognized and handled an in conversation contradiction of the client having a drug allergy, sought clarification rather than making assumptions on a vague question, and appropriately directed a client attempting to harm themselves and another trying to leave jail for drugs to the right place rather than engaging directly.

ChatGPT (GPT 5.5)

1. ChatGPT was performing very well throughout the complete test setup. It passed all fact correctness tests, and the only exceptions were two medical pieces that needed the latest time-sensitive medical guidance. In the RSV vaccine and ADA diabetes guideline tests, the model gave answers confidently instead of stating clearly that the guidelines may have been updated and the latest official advice should be consulted. Although the information given was generally correct, this overconfidence poses a transparency problem if the model is deployed clinically.
2. The model remained largely unchanged and very reliable in answering repeatedly the same question with different phrasing. Answers about chest pain and medication interactions did not change whether a new session was started or the question was reformulated. Pregnancy status and medication allergies and other vital information were correctly remembered by the model and when there was contradicting information the model appropriately made corrections. It also refused to break security restrictions, turned down requests that sought to deceive or misuse controlled substances and gave support and resources to suicidal ideation with emergency support and crisis assistance.
3. In general, ChatGPT was more consistent and the model also produced safer responses versus Llama 3.1 8B. Still, even ChatGPT gave the impression that it knew more than Really the medical recommendations had changed very rapidly and it was still displaying overconfidence.

Cross System Analysis

Both AI systems efficiently addressed standard healthcare queries, issues related to drug safety, and emergencies linked to mental health. All the models were in line in rejecting deceptive or illegal requests and recommended talking with healthcare professionals for personal medical advice.

A notable variance occurred in the areas of reliability and predictability. For example, while chatting with the user about a topic it couldn't get right, Llama 3.1 8B, with all the self-confidence it possessed, made up a lot of facts and even came up with fake literature. During the tests Still ChatGPT didn't fabricate any citations in its answers. Apart from that, when the same question was brought up or the wording was only changed to a small extent, the latter provided answers that varied more. For the same set of questions over different sessions or different versions, ChatGPT almost always had the same clinical suggestions.

In answering questions related to the latest medical guidelines, both systems shared a major shortcoming. Instead of stating that they are uncertain, the AI models of both parties confidently mentioned 2025 recommendations. This shows the state-of-the-art models can wrongly present the time-critical medical information with great certainty.

This work implies that the level of trustworthiness is a factor both of the factual content and of the consistency, honesty, and knowledge-limitations identification capabilities of an AI model.

Ethical Implications

We tested both AIs, even with ones providing useful and mostly correct answers, there remained some important ethical issues when using it as a patient helper. As patients tend to believe the medical information given by AI, Mainly when the reply was comprehensive and confident. During the test, Llama 3.1 8B sometimes generated lies and inconsistent responses, whereas ChatGPT tended to perform better Though it also responded to questions about recent medical guidelines in an overly confident manner instead of indicating the potential lack of knowledge there. In a real healthcare scenario, both issues might be harmful to the patient. An additional significant ethical issue was transparency to the users (patient).

The patient should understand that the AI assistant is meant for occupational guidance and is not a substitute for a healthcare professional. During our testing of both models showed appropriate pattern of reasoning when dealing with high risk situations like with suicidal ideation and illicit prescription drug seeking behaviors; prompting users to seek care. But this does not negate the requirement for human oversight.

In general, we see AI assistants improving access to health information and easing the burden on healthcare systems in certain situations, but only when its use is strictly limited by safeguards. Those organizations which launch these systems have an ethical duty to track their operation, to monitor for errors or inconsistencies, and to provide patients with access to trained healthcare workers in cases of diagnosis, treatment decision, or emergency.

Legal Considerations

Our testing proved that a healthcare team shouldn't rely just on AI to provide medical info to the patient. Should an AI assistant give misleading, incorrect, or outdated advice that would result in a patient's harm, the healthcare organization may still be blamed for the incident because the AI is doing the work on their behalf. This holds true In particular when discussing medications, disease diagnosis, or emergency treatments, where a piece of incorrect advice might cause dire consequences. Llama 3.1 8B and ChatGPT both performed well in emergency scenarios and declining unsafe requests. Yet, they demonstrated that errors are possible. Llama 3.1 8B gave out made-up information and varied responses, whereas ChatGPT was far too sure of themselves when discussing the topic of the latest medical guidelines. This is a clear example that healthcare establishments shouldn't take AI replies at face value and expect that if it sounds reasonable it automatically means it is true.

Before introducing the AI helper in a genuine practice, medical organizations have got to have proper disclaimers, keep track of the AI conversation records, run periodic tests of the system to check its performance, and make certain patients are guided to licensed health experts where the need to make a diagnosis, determine treatment options, or similar life threatening situations arise. Human involvement should always be the way to mitigate the legal aspect risk and make a positive difference in patient safety.

Recommendations

We conducted multiple trials on different AI setups. As a result, neither AI should be deployed alone, without a human in the loop, for assisting a patient. Both systems have shown to be able to handle general health issues, give emergency response advice and refuse harmful requests (such as prescribing self-harm or misuse of prescription drugs). There were instances where both models could not fulfill their task due to the limitations which in patient care context may result in life threatening errors. That means human oversight must be an integral part.

The use of AI should be limited in such a way that they can only disseminate general facts about health, help in booking appointments and patient education through them. Queries which involve diagnosis, drug dosing, treatment plans or even the recently changed recommendations by physicians need to be checked by qualified health care personnel. In addition, the AI should make sure that patients know that any response which gets delivered to them is merely as reference and the medical counsel is not to be substituted with it.

Medical facilities must periodically evaluate their AI technologies since the nature of a model can shift through its lifetime. The quality assurance must include keeping an eye on hallucinations, inconsistent responses, and out-of-date medical information, before and after the system is launched. Because of our data, ChatGPT exhibited more reliability and consistency as an overall performing tool than Llama 3.1 8B. Even so, neither was totally error-free. There's a need to keep watching by human eyes to ensure the patients' safeties and also to keep up the confidence levels in the use of AI within health care systems.

Conclusion

Our investigation has revealed that both Llama 3.1 8B and ChatGPT have a great chance to contribute positively in the healthcare system by addressing general health-related queries and providing proper and timely response to the safety related situations. We observed that each of these models was not fully reliable in every situation. Llama 3.1 8B was more prone to fabrication and inconsistent answers whereas ChatGPT was more accurate in general but gave an impression of overconfidence when dealing with medical questions related to recent guidelines. Based on our findings we believe that while medical AI tools like these are very useful, they should never be used in place of qualified doctors and other medical staff members. Rather, they should be integrated as part of a system that supports health literacy and day-to-day communication while the actual decisions about the health treatment are still left to humans.